

1. Standard Reference Textbooks & Deep-Dive Chapter Context

Subject Area	Standard Reference Textbook	Core Chapter Concept & Key Exam Takeaways
DC Circuits & Electrical	D. P. Kothari & I. J. Nagrath (McGraw Hill)	Ch 1 & 2: KVL ($\sum V=0$, Energy Conservation), KCL ($\sum I=0$, Charge Conservation), Voltage Divider $V_2=V_s \cdot R_2/(R_1+R_2)$, Current Divider $I_1=I_s \cdot R_2/(R_1+R_2)$, Thevenin (V_{th} , R_{th}), Norton (I_N , R_N), Max Power Transfer ($R_L=R_{th}$, $\eta=50\%$). Ch 3: RMS, Average, Form factor (1.11), Peak factor (1.414).
Electronic Devices & BJT	Robert L. Boylestad & Louis Nashelsky (Pearson)	Ch 1 & 2: P-N junction barrier (Si 0.7V, Ge 0.3V), Dynamic resistance $r_d=\Delta V_D/\Delta I_D=26mV/I_D$, Half-Wave ($\eta=40.6\%$, $\gamma=1.21$) vs Bridge Rectifier ($\eta=81.2\%$, $\gamma=0.482$, $PIV=V_m$), Zener diode regulator. Ch 3 & 4: BJT CE curves, $\alpha=I_C/I_E$, $\beta=I_C/I_B$, $\beta=\alpha/(1-\alpha)$, DC Load Line, Voltage Divider Bias.
Digital Logic & Design	M. Morris Mano & Michael D. Ciletti (Pearson)	Ch 2 & 3: De Morgan's Laws ($(A+B)'=A'B'$, $(AB)'=A'+B'$), Universal NAND (7400) & NOR (7402) realization, K-Maps. Ch 4: Half Adder ($Sum=A\oplus B$, $Carry=AB$) & Full Adder ($Sum=A\oplus B\oplus C_{in}$, $Cout=AB+BC_{in}+AC_{in}$ using 2 HAs + 1 OR gate IC 7432), 4-bit Ripple Carry Adder.
IoT & Microcontrollers	Raj Kamal (McGraw Hill)	Ch 1, 3 & 4: Arduino Uno ATmega328P (16MHz, 32KB Flash), 10-bit ADC (0-1023), Active-LOW IR sensor module (LM393 comparator), C++ code (pinMode, digitalRead, digitalWrite), Max GPIO pin current 40mA.

2. Curated YouTube Practical Video Hub & Circuit Demonstrations

Practical Experiment	Recommended Channel	Search Query / Video Reference	Core Practical Takeaway
DC Circuits & Theorems	Neso Academy	Neso Academy Network Theory KVL KCL	Mesh/Node analysis, Thevenin/Norton numericals.
Diodes & Rectifiers	All About Electronics	All About Electronics Half Wave and Full Wave Rectifier	P-N junction barrier, Bridge Rectifier waveforms.
BJT Transistors & Biasing	Neso Academy	Neso Academy BJT Transistor Biasing	CE characteristics, DC Load line, Voltage Divider Bias.
Universal Logic Gates	Neso Academy	Neso Academy Logic Gates Universal Gates	IC 7400 NAND & IC 7402 NOR gate realizations.
Half & Full Adders	Gate Smashers	Gate Smashers Half Adder and Full Adder	IC 7486 (XOR) + IC 7408 (AND) + IC 7432 (OR).
Arduino + IR Sensor	Circuit Digest	Arduino Uno IR Sensor Interfacing Circuit Digest	Active-LOW logic, Serial monitor debugging.
Viva Voce Preparation	Gate Smashers	Digital Electronics Viva Questions Gate Smashers	Top 50 examiner questions & troubleshooting.

3. High-Yield Theory Summary, Laboratory Rules & Precision Precautions

Mod 1 (DC Circuits & Laws): KVL ($\sum V=0$) is based on Conservation of Energy; KCL ($\sum I=0$) is based on Conservation of Charge. Voltage Divider: $V_2=V_s \cdot R_2/(R_1+R_2)$. Current Divider: $I_1=I_s \cdot R_2/(R_1+R_2)$. Thevenin V_{th} is open-circuit voltage across load; R_{th} is resistance with all sources deactivated (voltage sources shorted, current sources opened). Max power transfer occurs when $R_L = R_{th}$ (Efficiency $\eta = 50\%$).

Mod 2 & 3 (Semiconductor Diodes & BJT): Diode conducts in forward bias above Knee Voltage ($S_i = 0.7V$, $G_e = 0.3V$). Dynamic resistance $r_d = \Delta V_D/\Delta I_D = 26mV/I_D$. Half-Wave Rectifier: $\eta = 40.6\%$, $\gamma = 1.21$, $PIV = V_m$. Full-Wave Bridge: $\eta = 81.2\%$, $\gamma = 0.482$, $PIV = V_m$. BJT relations: $I_E = I_B + I_C$, $\alpha = I_C/I_E$, $\beta = I_C/I_B$, $\beta = \alpha/(1-\alpha)$. CE Active region acts as linear amplifier; Cutoff and Saturation act as digital switch. Voltage Divider Bias gives maximum Q-point stability.

Mod 4 & 5 (Digital Logic & Adders): NAND (IC 7400) and NOR (IC 7402) are universal. De Morgan's: $(A+B)' = A'B'$, $(AB)' = A'+B'$. Half Adder: $Sum = A\oplus B$ (IC 7486), $Carry = AB$ (IC 7408). Full Adder: $Sum = A\oplus B\oplus C_{in}$, $Cout = AB + BC_{in} + AC_{in}$ (using 2 Half Adders + 1 OR gate IC 7432). Standard DIP-14 pinout: Pin 14 = VCC (+5V), Pin 7 = GND. *IC 7402 NOR Exception: Pin 1 is Output, Pins 2 & 3 are Inputs!*

Mod 6 (Arduino Uno & IR Systems): ATmega328P (8-bit AVR RISC, 16MHz, 32KB Flash, 2KB SRAM). IR sensor module uses LM393 comparator with **Active-LOW logic**: outputs LOW (0V) on obstacle detection. Sketch logic: if (digitalRead(2) == LOW) digitalWrite(13, HIGH). Maximum GPIO pin current is 40mA (safe continuous $\leq 20mA$).

4. Top High-Frequency Viva Voce Scoring Blueprint

Q1: Why are NAND and NOR called Universal Gates? — <i>Ans:</i> Any Boolean function (AND, OR, NOT, XOR, XNOR) can be implemented solely using combinations of NAND or NOR gates.
Q2: What is the Knee / Cut-in voltage of Silicon vs Germanium Diodes? — <i>Ans:</i> Silicon cut-in is ~0.7V; Germanium cut-in is ~0.3V.
Q3: State KVL and KCL and their conservation laws. — <i>Ans:</i> KVL ($\sum V = 0$) represents Conservation of Energy; KCL ($\sum I = 0$) represents Conservation of Electric Charge.
Q4: How does an IR sensor output signal behave on obstacle detection? — <i>Ans:</i> Operates on Active-LOW logic; outputs Logic LOW (0V) when an obstacle is detected and HIGH (5V) when path is clear.
Q5: What is the condition and efficiency for Maximum Power Transfer? — <i>Ans:</i> Load resistance must equal Thevenin resistance ($R_L = R_{th}$); maximum efficiency is 50%.